

Tutorial/Assignment-2

Ques- Solve $\frac{\partial^2 y}{\partial x^2} + 6\frac{dy}{dx} + 9y = 6e^{3x} + 7e^{-2x} - \log 2$

$$\frac{\partial^2 y}{\partial x^2} + 6\frac{dy}{dx} + 9y = 0$$

$$m^2 + 6m + 9 = 0$$

$$C.F. = (C_1 + C_2 e^n) e^{-3x}$$

For particular solution,

$$6e^{3x} + 7e^{-2x} - \log 2$$

trial solution,

$$y_p = Ae^{3x} + Be^{-2x} + C$$

differentiating,

$$\frac{dy_p}{dx} = 3Ae^{3x} - 2Be^{-2x}$$

$$\frac{\partial^2 y_p}{\partial x^2} = 9Ae^{3x} + 4Be^{-2x}$$

Substituting,

$$(9Ae^{3x} + 4Be^{-2x}) + 6(3Ae^{3x} - 2Be^{-2x}) + 9(Ae^{3x} + Be^{-2x} + C) = 6e^{3x} + 7e^{-2x} - \log 2$$

Equating Coefficients,

$$36A = 6 \Rightarrow A = \frac{1}{6}$$

$$B = 7$$

$$9C = -\log 2 \Rightarrow C = -\frac{\log 2}{9}$$

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Particular solution = $\frac{1}{6} e^{3x} + 7e^{-2x} - \frac{\log 2}{9}$

General Solution,

$$y = (C_1 + C_2 x) e^{-3x} + \frac{1}{6} e^{3x} + 7e^{-2x} - \frac{\log 2}{9}$$

Ques 10 Solve the differential equation $\frac{\partial^3 y}{\partial x^3} = 7 \frac{\partial^2 y}{\partial x^2} + 10 \frac{\partial y}{\partial x}$

$$= e^{2x} \sin x$$

$$\frac{\partial^3 y}{\partial x^3} - 7 \frac{\partial^2 y}{\partial x^2} + 10 \frac{\partial y}{\partial x} = 0$$

$$m^3 - 7m^2 + 10m = 0$$

$$m^2 - 7m + 10 = (m-5)(m-2)$$

$$m=0, m=5, m=2$$

$$C.F. = C_1 + C_2 e^{5x} + C_3 e^{2x}$$

For particular solution,

trial solution, $\sin(x) e^{2x}$

differentiating,

$$y_g = e^{2x} (A \sin(x) + B \cos(x))$$

$$\frac{\partial y_g}{\partial x} = e^{2x} [A \cos(x) - B \sin(x) + 2(A \sin(x) + B \cos(x))]$$

$$\frac{\partial^2 y_g}{\partial x^2} = e^{2x} [(2A \cos(x) - 2B \sin(x) + 2(A \cos(x) - B \sin(x)) + 4(A \sin(x) + B \cos(x))]$$

$$\frac{\partial^3 y_p}{\partial x^3} = e^{2x} [(4A \cos(x) - 4B \sin(x)) + 4(2A \cos(x) - 2B \sin(x)) + 8(A \cos(x) - B \sin(x)) + 8(A \sin(x) + B \cos(x))]$$

Substituting and taking coefficients

$$A = -\frac{1}{10}, \quad B = \frac{1}{10}$$

$$y_p = e^{2x} \left(-\frac{1}{10} \sin(x) + \frac{1}{10} \cos(x) \right)$$

General solution,

$$y = C_1 + C_2 e^{5x} + C_3 e^{2x} + e^{2x} \left(-\frac{1}{10} \sin(x) + \frac{1}{10} \cos(x) \right)$$

Ques 3 Solve $\frac{\partial^2 y}{\partial x^2} - 3 \frac{\partial y}{\partial x} + 2y = \sin 2x$

$$\frac{\partial^2 y}{\partial x^2} - 3 \frac{\partial y}{\partial x} + 2y = 0$$

$$m^2 - 3m + 2 = 0$$

$$(m-1)(m-2) = 0$$

$$m = 1, 2$$

$$C.F. = C_1 e^x + C_2 e^{2x}$$

For particular solution,

$$\sin 2x = \frac{e^{ix} - e^{-ix}}{2i}$$

trial solution, $y_p = A e^x + B e^{-x}$

$$\frac{\partial y_p}{\partial x} = Ae^x - Be^{-x}$$

$$\frac{\partial^2 y_p}{\partial x^2} = Ae^x + Be^{-x}$$

Substituting,

$$(Ae^x + Be^{-x}) - 3(Ae^x - Be^{-x}) + 2(Ae^x + Be^{-x}) = \frac{e^x - e^{-x}}{2}$$

Equating Coefficients

$$A = 0, B = \frac{1}{12}$$

$$y_p = \frac{1}{12} e^{-x}$$

For general solⁿ,

$$y = C_1 e^x + C_2 e^{2x} + \frac{1}{12} e^{-x}$$

Ques- Find the directional derivative of $1/r$ in the direction $\vec{r}$ where $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$.

Directional derivative of scalar function $f(r)$ in the direction of a vector v is given by :-

$$D_v f = \nabla f \cdot \hat{v}$$

$$f(r) = 1/r$$

$$\nabla f(r) = f'(r) \frac{\vec{r}}{r}$$

$$f'(r) = -\frac{1}{r^2}$$

$$r = |\vec{r}|$$

$$\nabla f(r) = -\frac{1}{r^2} \cdot \frac{\vec{r}}{r} \Rightarrow -\frac{\vec{r}}{r^3}$$

directional derivative of $f(r)$ in direction of $\vec{r}$,

$$\nabla f(r) \cdot \frac{\vec{r}}{r} = -\frac{\vec{r}}{r^3} \cdot \frac{\vec{r}}{r} = -\frac{r^2}{r^4} = -\frac{1}{r^2}$$

$$= -\frac{1}{\sqrt{x^2+y^2+z^2}}$$

Ques 5 Find the angle b/w the surfaces $x^2+y^2+z^2=9$ & $z=x^2+y^2-3$ at the pt. $(2, -1, 2)$.

Soln Gradient of first surface, $x^2+y^2+z^2=9$,

$$\nabla f = \frac{\partial f}{\partial x} \hat{i} + \frac{\partial f}{\partial y} \hat{j} + \frac{\partial f}{\partial z} \hat{k}$$

$$\nabla f = 2x \hat{i} + 2y \hat{j} + 2z \hat{k}$$

at $(2, -1, 2)$,

$$\nabla f = 4\hat{i} - 2\hat{j} + 4\hat{k}$$

Gradient of second surface, $z-x^2-y^2+3=0$

$$\nabla g = \frac{\partial g}{\partial x} \hat{i} + \frac{\partial g}{\partial y} \hat{j} + \frac{\partial g}{\partial z} \hat{k}$$

$$\nabla g = -2x \hat{i} - 2y \hat{j} + \hat{k}$$

at $(2, -1, 2)$

$$\nabla g = -4\hat{i} + 2\hat{j} + \hat{k}$$

angle between the normal vectors,

$$\cos \theta = \frac{\nabla f \cdot \nabla g}{\|\nabla f\| \|\nabla g\|}$$

$$= \frac{4(-4) + (-2)(2) + 4(1)}{\sqrt{16+4+16} \sqrt{16+4+1}}$$

$$= \frac{-16}{\sqrt{21} \cdot 6}$$

$$= \frac{-8}{3\sqrt{21}}$$

$$\theta = \cos^{-1}\left(\frac{-8}{3\sqrt{21}}\right)$$